

MATHEMATICS-II

Unit-1: Mean Value Theorems & Successive Differentiation

Comprehensive Study Notes

TOPIC 1: ROLLE'S THEOREM

1.1 Statement of Rolle's Theorem

If a function $f(x)$ satisfies the following conditions:

1. **Continuous** in the closed interval $[a, b]$
2. **Differentiable** in the open interval (a, b)
3. $f(a) = f(b)$ (End points have equal values)

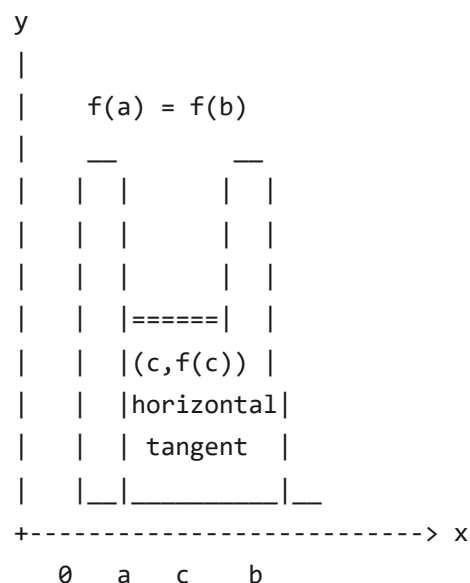
Then there exists **at least one** point $c \in (a, b)$ such that:

$$f'(c) = 0$$

1.2 Geometrical Significance

Geometrically, Rolle's theorem states that:

- The graph of the function has a **horizontal tangent** at some point between a and b
- The slope of the tangent at point c is **zero**
- The tangent is **parallel to the x-axis**



Interpretation:

- There is at least one point where the tangent is parallel to x-axis
- The function reaches a maximum or minimum at point c
- The slope $f'(c) = 0$

1.3 Applications of Rolle's Theorem

Application	Description
Existence of Roots	Prove that an equation has at least one root
Critical Points	Find points where derivative is zero
Verification	Check if theorem conditions are satisfied
Graph Analysis	Analyze behavior of functions

1.4 Solved Examples

Example 1: Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$

Solution:

Step 1: Check Conditions

1. Continuity: $f(x)$ is a polynomial, hence continuous on $[1, 3]$ ✓
2. Differentiability: $f(x)$ is differentiable on $(1, 3)$ ✓
3. End points:
 - $f(1) = 1 - 4 + 3 = 0$
 - $f(3) = 9 - 12 + 3 = 0$
 - $f(1) = f(3) = 0$ ✓

Step 2: Find c such that $f'(c) = 0$

- $f'(x) = 2x - 4$
- $f'(c) = 0 \Rightarrow 2c - 4 = 0$
- $2c = 4 \Rightarrow c = 2$

Step 3: Verify

- $c = 2 \in (1, 3)$ ✓

Conclusion: Rolle's theorem is verified and $c = 2$.

Example 2: Verify Rolle's Theorem for $f(x) = \sin x$ on $[0, \pi]$

Solution:

Step 1: Check Conditions

1. Continuity: $\sin x$ is continuous on $[0, \pi]$ ✓
2. Differentiability: $\sin x$ is differentiable on $(0, \pi)$ ✓
3. End points:
 - $f(0) = \sin 0 = 0$
 - $f(\pi) = \sin \pi = 0$
 - $f(0) = f(\pi) = 0$ ✓

Step 2: Find c such that $f'(c) = 0$

- $f'(x) = \cos x$
- $f'(c) = 0 \Rightarrow \cos c = 0$
- $c = \pi/2$

Step 3: Verify

- $c = \pi/2 \in (0, \pi)$ ✓

Conclusion: Rolle's theorem is verified and $c = \pi/2$.

Example 3: Verify Rolle's Theorem for $f(x) = x(x - 2)$ on $[0, 2]$

Solution:

Step 1: Check Conditions

1. Continuity: Polynomial, continuous ✓
2. Differentiability: Polynomial, differentiable ✓
3. End points:
 - $f(0) = 0(0-2) = 0$
 - $f(2) = 2(2-2) = 0$
 - $f(0) = f(2) = 0$ ✓

Step 2: Find c

- $f(x) = x^2 - 2x$
- $f'(x) = 2x - 2$
- $f'(c) = 0 \Rightarrow 2c - 2 = 0$
- $c = 1$

Step 3: Verify

$$- c = 1 \in (0, 2) \checkmark$$

Conclusion: Rolle's theorem is verified and $c = 1$.

Example 4: Find the value of c for $f(x) = e^x \sin x$ on $[0, \pi]$

Solution:

Step 1: Check Conditions

1. Continuity: $e^x \sin x$ is continuous $\checkmark$

2. Differentiability: Differentiable $\checkmark$

3. End points:

$$- f(0) = e^0 \sin 0 = 0$$

$$- f(\pi) = e^\pi \sin \pi = 0$$

$$- f(0) = f(\pi) = 0 \checkmark$$

Step 2: Find c

$$- f'(x) = e^x \sin x + e^x \cos x = e^x(\sin x + \cos x)$$

$$- f'(c) = 0 \Rightarrow e^c(\sin c + \cos c) = 0$$

$$- \text{Since } e^c \neq 0, \sin c + \cos c = 0$$

$$- \tan c = -1 \Rightarrow c = 3\pi/4$$

Step 3: Verify

$$- c = 3\pi/4 \in (0, \pi) \checkmark$$

Conclusion: $c = 3\pi/4$.

Example 5: Verify Rolle's Theorem for $f(x) = \log x$ on $[1, e]$

Solution:

Step 1: Check Conditions

1. Continuity: $\log x$ is continuous on $[1, e]$ $\checkmark$

2. Differentiability: $\log x$ is differentiable on $(1, e)$ $\checkmark$

3. End points:

$$- f(1) = \log 1 = 0$$

$$- f(e) = \log e = 1$$

$$- f(1) \neq f(e) \times$$

Conclusion: Rolle's theorem is NOT APPLICABLE because $f(1) \neq f(e)$.

Example 6: Verify Rolle's Theorem for $f(x) = |x|$ on $[-1, 1]$

Solution:

Step 1: Check Conditions

1. Continuity: $|x|$ is continuous on $[-1, 1]$ ✓
2. Differentiability: $|x|$ is NOT differentiable at $x = 0$ ✗

Conclusion: Rolle's theorem is NOT APPLICABLE because function is not differentiable

Example 7: Verify Rolle's Theorem for $f(x) = x^3 - 6x^2 + 11x - 6$ on $[1, 3]$

Solution:

Step 1: Check Conditions

1. Continuity: Polynomial, continuous ✓
2. Differentiability: Polynomial, differentiable ✓
3. End points:
 - $f(1) = 1 - 6 + 11 - 6 = 0$
 - $f(3) = 27 - 54 + 33 - 6 = 0$
 - $f(1) = f(3) = 0$ ✓

Step 2: Find c

- $f'(x) = 3x^2 - 12x + 11$
- $f'(c) = 0 \Rightarrow 3c^2 - 12c + 11 = 0$
- $c = [12 \pm \sqrt{(144 - 132)}] / 6 = [12 \pm \sqrt{12}] / 6$
- $c = [12 \pm 2\sqrt{3}] / 6 = 2 \pm \sqrt{3}/3$
- $c_{\text{sub}1} = 2 + \sqrt{3}/3 \approx 2.577$, $c_{\text{sub}2} = 2 - \sqrt{3}/3 \approx 1.423$

Step 3: Verify

- Both $c_{\text{sub}1}$, $c_{\text{sub}2} \in (1, 3)$ ✓

Conclusion: Rolle's theorem is verified and $c = 2 \pm \sqrt{3}/3$.

Example 8: Show that $x^3 - 3x + 1 = 0$ has at least one root between 0 and 1

Solution:

Step 1: Define $f(x) = x^3 - 3x + 1$

Step 2: Use Intermediate Value Theorem

- $f(0) = 0 - 0 + 1 = 1$
- $f(1) = 1 - 3 + 1 = -1$
- Since $f(0)$ and $f(1)$ have opposite signs, there is at least one root in $(0, 1)$.

Conclusion: Equation has at least one root between 0 and 1.

1.5 Quick Practice Questions

Q.No.	Question
Q1	Verify Rolle's theorem for $f(x) = x^2 + 2x$ on $[-2, 0]$
Q2	Verify Rolle's theorem for $f(x) = \cos x$ on $[-\pi/2, \pi/2]$
Q3	Verify Rolle's theorem for $f(x) = x(x-1)(x-2)$ on $[0, 2]$
Q4	Find c for $f(x) = e^x \cos x$ on $[0, \pi/2]$
Q5	Check if Rolle's theorem is applicable to $f(x) = x^{2/3}$ on $[-1, 1]$

TOPIC 2: LAGRANGE'S MEAN VALUE THEOREM

2.1 Statement of Lagrange's MVT

If a function $f(x)$ satisfies the following conditions:

1. **Continuous** in the closed interval $[a, b]$
2. **Differentiable** in the open interval (a, b)

Then there exists **at least one** point $c \in (a, b)$ such that:

$$f'(c) = [f(b) - f(a)] / (b - a)$$

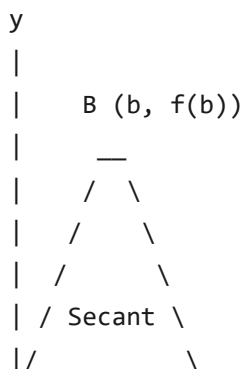
OR

$$f(b) - f(a) = (b - a) \cdot f'(c)$$

2.2 Geometrical Significance

Geometrically, Lagrange's MVT states that:

- There exists at least one point on the curve where the **tangent is parallel** to the **secant line** joining the endpoints
- The slope of the tangent at point c equals the slope of the secant line



A (a, f(a))

+-----> x

0 a c b

Interpretation:

- The average rate of change equals the instantaneous rate of change at some point
- The chord AB is parallel to the tangent at c
- $f'(c)$ represents the instantaneous rate of change

2.3 Applications of Lagrange's MVT

Application	Description
Proving Inequalities	Used to prove various mathematical inequalities
Error Estimation	Used in numerical methods
Taylor's Theorem	Foundation for Taylor series
Rolle's Theorem	Generalization of Rolle's theorem

2.4 Solved Examples

Example 1: Verify Lagrange's MVT for $f(x) = x^2$ on $[1, 4]$

Solution:

Step 1: Check Conditions

1. Continuity: Polynomial, continuous on $[1, 4]$ ✓
2. Differentiability: Polynomial, differentiable on $(1, 4)$ ✓

Step 2: Find c

- $f(1) = 1$
- $f(4) = 16$
- $[f(4) - f(1)] / (4 - 1) = (16 - 1) / 3 = 15 / 3 = 5$
- $f'(x) = 2x$
- $f'(c) = 5 \Rightarrow 2c = 5 \Rightarrow c = 5/2 = 2.5$

Step 3: Verify

$$- c = 2.5 \in (1, 4) \checkmark$$

Conclusion: Lagrange's MVT is verified and $c = 2.5$.

Example 2: Verify Lagrange's MVT for $f(x) = \log x$ on $[1, e]$

Solution:

Step 1: Check Conditions

1. Continuity: $\log x$ is continuous on $[1, e] \checkmark$
2. Differentiability: $\log x$ is differentiable on $(1, e) \checkmark$

Step 2: Find c

$$\begin{aligned} - f(1) &= \log 1 = 0 \\ - f(e) &= \log e = 1 \\ - [f(e) - f(1)] / (e - 1) &= 1 / (e - 1) \\ - f'(x) &= 1/x \\ - f'(c) &= 1/c = 1 / (e - 1) \Rightarrow c = e - 1 \end{aligned}$$

Step 3: Verify

$$- c = e - 1 \approx 1.718 \in (1, e) \checkmark$$

Conclusion: Lagrange's MVT is verified and $c = e - 1$.

Example 3: Find c for $f(x) = x^3 - 2x^2 + 3x + 5$ on $[1, 4]$

Solution:

Step 1: Check Conditions

1. Continuity: Polynomial $\checkmark$
2. Differentiability: Polynomial $\checkmark$

Step 2: Find c

$$\begin{aligned} - f(1) &= 1 - 2 + 3 + 5 = 7 \\ - f(4) &= 64 - 32 + 12 + 5 = 49 \\ - [f(4) - f(1)] / (4 - 1) &= (49 - 7) / 3 = 42 / 3 = 14 \\ - f'(x) &= 3x^2 - 4x + 3 \\ - f'(c) &= 14 \Rightarrow 3c^2 - 4c + 3 = 14 \\ - 3c^2 - 4c - 11 &= 0 \\ - c &= [4 \pm \sqrt{(16 + 132)}] / 6 = [4 \pm \sqrt{148}] / 6 \end{aligned}$$

- $c = [4 \pm 2\sqrt{37}]/6 = (2 \pm \sqrt{37})/3$
- $c_{\text{sub1}} = (2+\sqrt{37})/3 \approx 2.694$ (accepted)
- $c_{\text{sub2}} = (2-\sqrt{37})/3 \approx -1.361$ (rejected)

Step 3: Verify

- $c = (2+\sqrt{37})/3 \approx 2.694 \in (1, 4) \checkmark$

Conclusion: $c = (2+\sqrt{37})/3$.

Example 4: Using Lagrange's MVT, prove that $x/(1+x) < \log(1+x) < x$ for $x > 0$

Solution:

Step 1: Apply Lagrange's MVT

- Let $f(t) = \log(1+t)$ on $[0, x]$
- $f'(t) = 1/(1+t)$
- By Lagrange's MVT: $[f(x)-f(0)]/(x-0) = f'(c)$ for some $c \in (0, x)$
- $[\log(1+x) - \log 1]/x = 1/(1+c)$
- $\log(1+x)/x = 1/(1+c)$

Step 2: Since $0 < c < x$

- $1 < 1+c < 1+x$
- $1/(1+x) < 1/(1+c) < 1$
- $1/(1+x) < \log(1+x)/x < 1$
- $x/(1+x) < \log(1+x) < x \checkmark$

Conclusion: Proved.

Example 5: Verify Lagrange's MVT for $f(x) = e^x$ on $[0, 1]$

Solution:

Step 1: Check Conditions

1. Continuity: e^x is continuous $\checkmark$
2. Differentiability: e^x is differentiable $\checkmark$

Step 2: Find c

- $f(0) = e^0 = 1$
- $f(1) = e$
- $[f(1)-f(0)]/(1-0) = e - 1$
- $f'(x) = e^x$

- $f'(c) = e^c = e - 1$
- $c = \log(e - 1)$

Step 3: Verify

- $c = \log(e-1) \approx 0.541 \in (0, 1) \checkmark$

Conclusion: $c = \log(e - 1)$.

Example 6: Find c for $f(x) = \sin x$ on $[0, \pi/2]$

Solution:

Step 1: Check Conditions

1. Continuity: $\sin x$ is continuous $\checkmark$
2. Differentiability: $\sin x$ is differentiable $\checkmark$

Step 2: Find c

- $f(0) = 0$
- $f(\pi/2) = 1$
- $[f(\pi/2) - f(0)] / (\pi/2 - 0) = 1 / (\pi/2) = 2/\pi$
- $f'(x) = \cos x$
- $f'(c) = \cos c = 2/\pi$
- $c = \cos^{-1}(2/\pi)$

Step 3: Verify

- $c = \cos^{-1}(2/\pi) \approx 0.881 \in (0, \pi/2) \checkmark$

Conclusion: $c = \cos^{-1}(2/\pi)$.

Example 7: Verify Lagrange's MVT for $f(x) = \sqrt{x}$ on $[1, 4]$

Solution:

Step 1: Check Conditions

1. Continuity: $\sqrt{x}$ is continuous $\checkmark$
2. Differentiability: $\sqrt{x}$ is differentiable on $(1, 4) \checkmark$

Step 2: Find c

- $f(1) = 1$
- $f(4) = 2$
- $[f(4) - f(1)] / (4 - 1) = (2 - 1) / 3 = 1/3$

- $f'(x) = 1/(2\sqrt{x})$
- $f'(c) = 1/(2\sqrt{c}) = 1/3$
- $2\sqrt{c} = 3 \Rightarrow \sqrt{c} = 3/2 \Rightarrow c = 9/4 = 2.25$

Step 3: Verify

- $c = 2.25 \in (1, 4) \checkmark$

Conclusion: $c = 9/4$.

2.5 Quick Practice Questions

Q.No. **Question**

Q1	Verify Lagrange's MVT for $f(x) = x^2 - 2x + 3$ on $[1, 3]$
Q2	Find c for $f(x) = \sin x$ on $[0, \pi]$
Q3	Verify Lagrange's MVT for $f(x) = \sqrt{x}$ on $[4, 9]$
Q4	Find c for $f(x) = e^x$ on $[0, 2]$
Q5	Prove $x/(1+x^2) < \tan^{-1}x < x$ for $x > 0$ using Lagrange's MVT

TOPIC 3: CAUCHY'S MEAN VALUE THEOREM

3.1 Statement of Cauchy's MVT

If two functions $f(x)$ and $g(x)$ satisfy the following conditions:

1. **Continuous** in the closed interval $[a, b]$
2. **Differentiable** in the open interval (a, b)
3. $g'(x) \neq 0$ for all $x \in (a, b)$

Then there exists **at least one** point $c \in (a, b)$ such that:

$$[f(b) - f(a)] / [g(b) - g(a)] = f'(c) / g'(c)$$

3.2 Geometrical Significance

Geometrically, Cauchy's MVT states that:

- The ratio of the changes in f and g equals the ratio of their derivatives at some point c
- It is a generalized form of Lagrange's MVT (when $g(x) = x$)

3.3 Relation with Lagrange's MVT

If we take $g(x) = x$, then:

- $g(b) - g(a) = b - a$
- $g'(c) = 1$

Thus, Cauchy's MVT reduces to Lagrange's MVT:

$$[f(b) - f(a)] / (b - a) = f'(c)$$

3.4 Applications of Cauchy's MVT

Application	Description
Indeterminate Forms	Foundation for L'Hôpital's Rule
Parametric Equations	Analysis of parametric curves
Generalization	Generalizes Lagrange's MVT
Proving Results	Used to prove various theorems

3.5 Solved Examples

Example 1: Verify Cauchy's MVT for $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$

Solution:

Step 1: Check Conditions

1. Continuity: Both continuous ✓
2. Differentiability: Both differentiable ✓
3. $g'(x)$: $g'(x) = 3x^2 \neq 0$ for $x \in (1, 2)$ ✓

Step 2: Find c

- $f(1) = 1, f(2) = 4$
- $g(1) = 1, g(2) = 8$
- $[f(2)-f(1)]/[g(2)-g(1)] = (4-1)/(8-1) = 3/7$
- $f'(x) = 2x, g'(x) = 3x^2$
- $f'(c)/g'(c) = 2c/(3c^2) = 2/(3c)$
- $2/(3c) = 3/7 \Rightarrow 14 = 9c \Rightarrow c = 14/9$

Step 3: Verify

- $c = 14/9 \approx 1.556 \in (1, 2)$ ✓

Conclusion: Cauchy's MVT is verified and $c = 14/9$.

Example 2: Find c for $f(x) = \sin x$ and $g(x) = \cos x$ on $[0, \pi/2]$

Solution:

Step 1: Check Conditions

1. Continuity: Both continuous ✓
2. Differentiability: Both differentiable ✓
3. $g'(x)$: $g'(x) = -\sin x \neq 0$ for $x \in (0, \pi/2)$ ✓

Step 2: Find c

- $f(0) = 0, f(\pi/2) = 1$
- $g(0) = 1, g(\pi/2) = 0$
- $[f(\pi/2)-f(0)]/[g(\pi/2)-g(0)] = (1-0)/(0-1) = -1$
- $f'(x) = \cos x, g'(x) = -\sin x$
- $f'(c)/g'(c) = \cos c/(-\sin c) = -\cot c$
- $-\cot c = -1 \Rightarrow \cot c = 1 \Rightarrow c = \pi/4$

Step 3: Verify

- $c = \pi/4 \in (0, \pi/2)$ ✓

Conclusion: $c = \pi/4$.

Example 3: Verify Cauchy's MVT for $f(x) = x^3$ and $g(x) = x^2$ on $[1, 3]$

Solution:

Step 1: Check Conditions

1. Continuity: Both continuous ✓
2. Differentiability: Both differentiable ✓
3. $g'(x)$: $g'(x) = 2x \neq 0$ for $x \in (1, 3)$ ✓

Step 2: Find c

- $f(1) = 1, f(3) = 27$
- $g(1) = 1, g(3) = 9$
- $[f(3)-f(1)]/[g(3)-g(1)] = (27-1)/(9-1) = 26/8 = 13/4$
- $f'(x) = 3x^2, g'(x) = 2x$
- $f'(c)/g'(c) = 3c^2/(2c) = 3c/2$
- $3c/2 = 13/4 \Rightarrow 12c = 26 \Rightarrow c = 13/6$

Step 3: Verify

- $c = 13/6 \approx 2.167 \in (1, 3)$ ✓

Conclusion: $c = 13/6$.

Example 4: Find c for $f(x) = e^x$ and $g(x) = e^{-x}$ on $[0, 1]$

Solution:

Step 1: Check Conditions

1. Continuity: Both continuous ✓
2. Differentiability: Both differentiable ✓
3. $g'(x)$: $g'(x) = -e^{-x} \neq 0$ ✓

Step 2: Find c

- $f(0) = 1, f(1) = e$
- $g(0) = 1, g(1) = e^{-1}$
- $[f(1)-f(0)]/[g(1)-g(0)] = (e-1)/(e^{-1}-1) = (e-1)/((1-e)/e) = -e$
- $f'(x) = e^x, g'(x) = -e^{-x}$
- $f'(c)/g'(c) = e^c/(-e^{-c}) = -e^{2c}$
- $-e^{2c} = -e \Rightarrow e^{2c} = e \Rightarrow 2c = 1 \Rightarrow c = 1/2$

Step 3: Verify

- $c = 1/2 \in (0, 1)$ ✓

Conclusion: $c = 1/2$.

Example 5: Verify Cauchy's MVT for $f(x) = \log x$ and $g(x) = 1/x$ on $[1, e]$

Solution:

Step 1: Check Conditions

1. Continuity: Both continuous ✓
2. Differentiability: Both differentiable ✓
3. $g'(x)$: $g'(x) = -1/x^2 \neq 0$ ✓

Step 2: Find c

- $f(1) = 0, f(e) = 1$
- $g(1) = 1, g(e) = 1/e$
- $[f(e)-f(1)]/[g(e)-g(1)] = 1/(1/e-1) = 1/((1-e)/e) = e/(1-e)$
- $f'(x) = 1/x, g'(x) = -1/x^2$
- $f'(c)/g'(c) = (1/c)/(-1/c^2) = -c$
- $-c = e/(1-e) \Rightarrow c = e/(e-1)$

Step 3: Verify

- $c = e/(e-1) \approx 1.582 \in (1, e)$ ✓

Conclusion: $c = e/(e-1)$.

3.6 Quick Practice Questions

Q.No.	Question
Q1	Verify Cauchy's MVT for $f(x) = x^2$ and $g(x) = \sqrt{x}$ on $[1, 4]$
Q2	Find c for $f(x) = \sin x$ and $g(x) = \cos x$ on $[0, \pi]$
Q3	Verify Cauchy's MVT for $f(x) = x^3$ and $g(x) = x$ on $[1, 2]$
Q4	Find c for $f(x) = e^x$ and $g(x) = x^2$ on $[1, 2]$
Q5	Verify Cauchy's MVT for $f(x) = \tan x$ and $g(x) = \sec x$ on $[0, \pi/4]$

TOPIC 4: SUCCESSIVE DIFFERENTIATION

4.1 Definition

Successive Differentiation is the process of differentiating a function repeatedly.

- **First derivative:** dy/dx or y_1
- **Second derivative:** d^2y/dx^2 or y_2
- **nth derivative:** d^ny/dx^n or y_n

4.2 Notation

Order	Notation 1	Notation 2
0th	y	$f(x)$
1st	y_1 or y'	$f'(x)$
2nd	y_2 or y''	$f''(x)$
3rd	y_3 or y'''	$f'''(x)$
nth	y_n	$f^{(n)}(x)$

4.3 nth Derivatives of Standard Functions

Function	nth Derivative
x^m	$m(m-1)(m-2)\dots(m-n+1) \cdot x^{m-n}$
e^{ax}	$a^n e^{ax}$
a^x	$a^x (\log a)^n$

$\sin(ax+b)$	$a^n \sin(ax+b + n\pi/2)$
$\cos(ax+b)$	$a^n \cos(ax+b + n\pi/2)$
$\log(ax+b)$	$(-1)^{n-1} (n-1)! a^n / (ax+b)^n$
$1/(ax+b)$	$(-1)^n n! a^n / (ax+b)^{n+1}$
$e^{ax} \sin(bx+c)$	$(a^2+b^2)^{n/2} e^{ax} \sin(bx+c + n \tan^{-1}(b/a))$
$e^{ax} \cos(bx+c)$	$(a^2+b^2)^{n/2} e^{ax} \cos(bx+c + n \tan^{-1}(b/a))$

4.4 Solved Examples

Example 1: Find the n th derivative of $y = x^n$

$$\begin{aligned}
 y &= x^n \\
 y_{\text{sub}1} &= nx^{n-1} \\
 y_{\text{sub}2} &= n(n-1)x^{n-2} \\
 y_{\text{sub}3} &= n(n-1)(n-2)x^{n-3} \\
 &\dots \\
 y_n &= n(n-1)(n-2)\dots(3)(2)(1)x^0 = n!
 \end{aligned}$$

Conclusion: $y_n = n!$

Example 2: Find the n th derivative of $y = e^{ax}$

$$\begin{aligned}
 y &= e^{ax} \\
 y_{\text{sub}1} &= a e^{ax} \\
 y_{\text{sub}2} &= a^2 e^{ax} \\
 y_{\text{sub}3} &= a^3 e^{ax} \\
 &\dots \\
 y_n &= a^n e^{ax}
 \end{aligned}$$

Conclusion: $y_n = a^n e^{ax}$.

Example 3: Find the nth derivative of $y = \sin(ax+b)$

$$\begin{aligned} y &= \sin(ax+b) \\ y_{\text{sub}1} &= a \cos(ax+b) = a \sin(ax+b + \pi/2) \\ y_{\text{sub}2} &= -a^2 \sin(ax+b) = a^2 \sin(ax+b + \pi) \\ y_{\text{sub}3} &= -a^3 \cos(ax+b) = a^3 \sin(ax+b + 3\pi/2) \\ &\dots \\ y_n &= a^n \sin(ax+b + n\pi/2) \end{aligned}$$

Conclusion: $y_n = a^n \sin(ax+b + n\pi/2)$.

Example 4: Find the nth derivative of $y = \log(ax+b)$

$$\begin{aligned} y &= \log(ax+b) \\ y_{\text{sub}1} &= a/(ax+b) \\ y_{\text{sub}2} &= -a^2/(ax+b)^2 \\ y_{\text{sub}3} &= 2! a^3/(ax+b)^3 \\ y_{\text{sub}4} &= -3! a^4/(ax+b)^4 \\ &\dots \\ y_n &= (-1)^{n-1} (n-1)! a^n / (ax+b)^n \end{aligned}$$

Example 5: Find the nth derivative of $y = e^x \sin x$

$$\begin{aligned} y &= e^x \sin x \\ \text{Here } a &= 1, b = 1, c = 0 \\ y_n &= (1^2+1^2)^{n/2} e^x \sin(x + n \tan^{-1}(1)) \\ &= 2^{n/2} e^x \sin(x + n\pi/4) \end{aligned}$$

Conclusion: $y_n = 2^{n/2} e^x \sin(x + n\pi/4)$.

Example 6: Find the nth derivative of $y = x^2 \sin x$

$$\begin{aligned} &\text{Using product rule for nth derivative:} \\ \text{Let } u &= x^2, v = \sin x \\ u_{\text{sub}0} &= x^2, u_{\text{sub}1} = 2x, u_{\text{sub}2} = 2, u_{\text{sub}3} = 0, \dots \\ v_n &= \sin(x + n\pi/2) \end{aligned}$$

By Leibnitz's theorem:

$$y_n = \frac{d^n}{dx^n} (x^2 \sin(x + n\pi/2)) = x^2 \sin(x + n\pi/2) + n(2x) \sin(x + (n-1)\pi/2) + \frac{n(n-1)}{2!} \sin(x + (n-2)\pi/2)$$

Conclusion: $y_n = x^2 \sin(x + n\pi/2) + 2nx \sin(x + (n-1)\pi/2) + n(n-1) \sin(x + (n-2)\pi/2)$.



TOPIC 5: LEIBNITZ'S THEOREM

5.1 Statement

If u and v are two functions of x , then the n th derivative of their product is:

$$(uv)_n = {}^nC_0 u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_r u_{n-r} v_r + \dots + u v_n$$

OR

$$(uv)_n = \sum_{r=0}^n {}^nC_r u_{n-r} v_r$$

where:

- u_n = n th derivative of u
- v_n = n th derivative of v
- ${}^nC_r = n! / [r!(n-r)!]$

5.2 Applications of Leibnitz's Theorem

Application	Description
Product of Functions	Find n th derivative of product of two functions
Finding Derivatives	Simplify complex differentiation
Formation of Equations	Form differential equations
Verification	Verify results of successive differentiation

5.3 Solved Examples

Example 1: Find y_n of $y = x^2 \log x$ using Leibnitz's theorem

Step 1: Identify u and v

Let $u = x^2$, $v = \log x$

$u_{\text{sub}0} = x^2$, $u_{\text{sub}1} = 2x$, $u_{\text{sub}2} = 2$, $u_{\text{sub}3} = 0$, ...

$v_{\text{sub}1} = 1/x$, $v_{\text{sub}2} = -1/x^2$, $v_{\text{sub}3} = 2/x^3$, $v_{\text{sub}4} = -6/x^4$, ...

$v_n = (-1)^{n-1}(n-1)!/x^n$

Step 2: Apply Leibnitz's theorem

$y_n = \sum_{k=0}^n C_{\text{sub}0} u_{\text{sub}0} v_n + \sum_{k=1}^n C_{\text{sub}1} u_{\text{sub}1} v_{n-1} + \sum_{k=2}^n C_{\text{sub}2} u_{\text{sub}2} v_{n-2} + \dots$

$$= x^2 \cdot (-1)^{n-1}(n-1)!/x^n + n(2x) \cdot (-1)^{n-2}(n-2)!/x^{n-1} + [n(n-1)/2](2) \cdot (-1)^{n-3}(n-3)!/x^{n-2}$$

Step 3: Simplify

$$y_n = (-1)^{n-1}(n-1)!/x^{n-2} + 2n(-1)^{n-2}(n-2)!/x^{n-2} + n(n-1)(-1)^{n-3}(n-3)!/x^{n-2}$$

$$y_n = [(-1)^{n-1}(n-1)! + 2n(-1)^{n-2}(n-2)! + n(n-1)(-1)^{n-3}(n-3)!] / x^{n-2}$$

Example 2: Find y_n of $y = x e^x$ using Leibnitz's theorem

Step 1: Identify u and v

Let $u = x$, $v = e^x$

$u_{\text{sub}0} = x$, $u_{\text{sub}1} = 1$, $u_{\text{sub}2} = 0$, ...

$v_n = e^x$

Step 2: Apply Leibnitz's theorem

$$y_n = \sum_{k=0}^n C_{\text{sub}0} u_{\text{sub}0} v_n + \sum_{k=1}^n C_{\text{sub}1} u_{\text{sub}1} v_{n-1} + 0 \\ = x e^x + n(1) e^x$$

$$y_n = e^x(x + n)$$

Example 3: Find y_n of $y = x^2 e^{2x}$

Step 1: Identify u and v

Let $u = x^2$, $v = e^{2x}$

$u_{\text{sub}0} = x^2$, $u_{\text{sub}1} = 2x$, $u_{\text{sub}2} = 2$, $u_{\text{sub}3} = 0$, ...

$v_n = 2^n e^{2x}$

Step 2: Apply Leibnitz's theorem

$y_n = \sum_{k=0}^n C_{\text{sub}0}^n u_{\text{sub}k} v_{n-k} + \sum_{k=1}^n C_{\text{sub}1}^n u_{\text{sub}k} v_{n-k} + \sum_{k=2}^n C_{\text{sub}2}^n u_{\text{sub}k} v_{n-k} + \dots$
 $= x^2(2^n e^{2x}) + n(2x)(2^{n-1} e^{2x}) + [n(n-1)/2](2)(2^{n-2} e^{2x})$

Step 3: Simplify

$y_n = 2^n e^{2x} [x^2 + nx + n(n-1)/4]$

$$y_n = 2^n e^{2x} [x^2 + nx + n(n-1)/4]$$

Example 4: Find y_n of $y = e^x \log x$

Step 1: Identify u and v

Let $u = e^x$, $v = \log x$

$u_n = e^x$

$v_n = (-1)^{n-1} (n-1)! / x^n$

Step 2: Apply Leibnitz's theorem

$y_n = \sum_{k=0}^n C_{\text{sub}0}^n u_{\text{sub}k} v_{n-k} + \sum_{k=1}^n C_{\text{sub}1}^n u_{\text{sub}k} v_{n-k} + \sum_{k=2}^n C_{\text{sub}2}^n u_{\text{sub}k} v_{n-k} + \dots$
 $= e^x v_n + n e^x v_{n-1} + [n(n-1)/2] e^x v_{n-2} + \dots + e^x v_{\text{sub}0}$

$$y_n = e^x [(-1)^{n-1} (n-1)! / x^n + n(-1)^{n-2} (n-2)! / x^{n-1} + \dots + \log x]$$

Example 5: If $y = \sin^{-1} x$, prove that $(1-x^2)y_2 - xy_1 = 0$

Step 1: Given $y = \sin^{-1} x$

$y_{\text{sub}1} = 1/\sqrt{1-x^2}$

Step 2: Square both sides

$y_{\text{sub}1}^2 = 1/(1-x^2)$

$(1-x^2)y_{\text{sub}1}^2 = 1$

Step 3: Differentiate with respect to x

$$(1-x^2)y'' + (-2x)y' = 0$$

$$(1-x^2)y'' - xy' = 0$$

$$(1-x^2)y'' - xy' = 0 \quad \checkmark$$

Example 6: If $y = \tan^{-1}x$, prove that $(1+x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$

Step 1: Given $y = \tan^{-1}x$

$$y' = 1/(1+x^2)$$

$$(1+x^2)y' = 1$$

Step 2: Differentiate with respect to x

$$(1+x^2)y'' + 2xy' = 0$$

Step 3: Apply Leibnitz's theorem with n

Differentiate the equation n times:

$$(1+x^2)y_{n+2} + 2xy_{n+1} + n(n+1)y_n = 0$$

$$(1+x^2)y_{n+2} + 2nx y_{n+1} + n(n-1)y_n + 2x y_{n+1} + 2n y_n = 0$$

$$(1+x^2)y_{n+2} + 2(n+1)x y_{n+1} + n(n+1)y_n = 0$$

$$(1+x^2)y_{n+2} + 2(n+1)x y_{n+1} + n(n+1)y_n = 0$$

5.4 Quick Practice Questions

Q.No.	Question
Q1	Find y_n of $y = x^3 \log x$
Q2	Find y_n of $y = x e^{2x}$
Q3	Find y_n of $y = x^2 \sin x$
Q4	If $y = \cos^{-1}x$, prove $(1-x^2)y_2 - xy_1 = 0$
Q5	If $y = \tan^{-1}x$, prove $(1+x^2)y_2 + 2xy_1 = 0$

Q6

If $y = \sin^{-1}x$, prove $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$

UNIT-I: HPU IMPORTANT QUESTIONS

Short Answer Questions (2-5 Marks)

1. State Rolle's theorem.
2. State Lagrange's mean value theorem.
3. State Cauchy's mean value theorem.
4. State Leibnitz's theorem.
5. Write the n th derivative of e^{ax} .
6. Write the n th derivative of $\sin(ax+b)$.
7. Write the n th derivative of $\log(ax+b)$.
8. What is the geometrical significance of Rolle's theorem?
9. What is the geometrical significance of Lagrange's MVT?
10. What is the relation between Rolle's theorem and Lagrange's MVT?

Long Answer Questions (10 Marks)

1. State and prove Rolle's theorem.
2. State and prove Lagrange's mean value theorem.
3. State and prove Cauchy's mean value theorem.
4. State and prove Leibnitz's theorem.
5. Explain the geometrical significance of mean value theorems.
6. Verify Rolle's theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$.
7. Verify Lagrange's MVT for $f(x) = x^2 + 2x$ on $[1, 3]$.
8. Verify Cauchy's MVT for $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$.
9. Find the n th derivative of $y = e^{ax} \sin(bx+c)$.
10. Find y_n of $y = x^2 \log x$ using Leibnitz's theorem.

Objective Questions (1 Mark)

1. Rolle's theorem requires:
(a) $f(a) = f(b)$ (b) $f(a) \neq f(b)$ (c) $f(a) = 0$ (d) $f(b) = 0$

Answer: (a)

2. Lagrange's MVT gives:

- (a) $f'(c) = 0$ (b) $f'(c) = (f(b)-f(a))/(b-a)$ (c) $f'(c) = f(b) - f(a)$ (d) None

Answer: (b)

3. The n th derivative of e^{ax} is:

- (a) $a^n e^{ax}$ (b) e^{ax} (c) $a e^{ax}$ (d) 0

Answer: (a)

4. The n th derivative of $\sin x$ is:

- (a) $\sin(x + n\pi/2)$ (b) $\cos(x + n\pi/2)$ (c) $\sin nx$ (d) $\cos nx$

Answer: (a)

5. Leibnitz's theorem is used for:

- (a) n th derivative of product (b) n th derivative of sum (c) n th derivative of quotient (d) None

Answer: (a)

QUICK REVISION

Mean Value Theorems Summary

Theorem	Conditions	Result
Rolle's Theorem	Continuous on $[a,b]$, Differentiable on (a,b) , $f(a)=f(b)$	$\exists c \in (a,b): f'(c) = 0$
Lagrange's MVT	Continuous on $[a,b]$, Differentiable on (a,b)	$\exists c \in (a,b): f'(c) = (f(b)-f(a))/(b-a)$
Cauchy's MVT	f,g continuous on $[a,b]$, differentiable on (a,b) , $g'(x) \neq 0$	$\exists c \in (a,b): (f(b)-f(a))/(g(b)-g(a)) = f'(c)/g'(c)$

nth Derivatives Summary

Function	nth Derivative
x^m	$m(m-1)\dots(m-n+1)x^{m-n}$
e^{ax}	$a^n e^{ax}$
$\sin(ax+b)$	$a^n \sin(ax+b + n\pi/2)$
$\cos(ax+b)$	$a^n \cos(ax+b + n\pi/2)$
$\log(ax+b)$	$(-1)^{n-1}(n-1)!a^n/(ax+b)^n$

Leibnitz's Theorem

$$(uv)_n = \sum_{r=0}^n {}^n C_r u_{n-r} v_r$$

End of Unit-1 Notes